

Agenda:

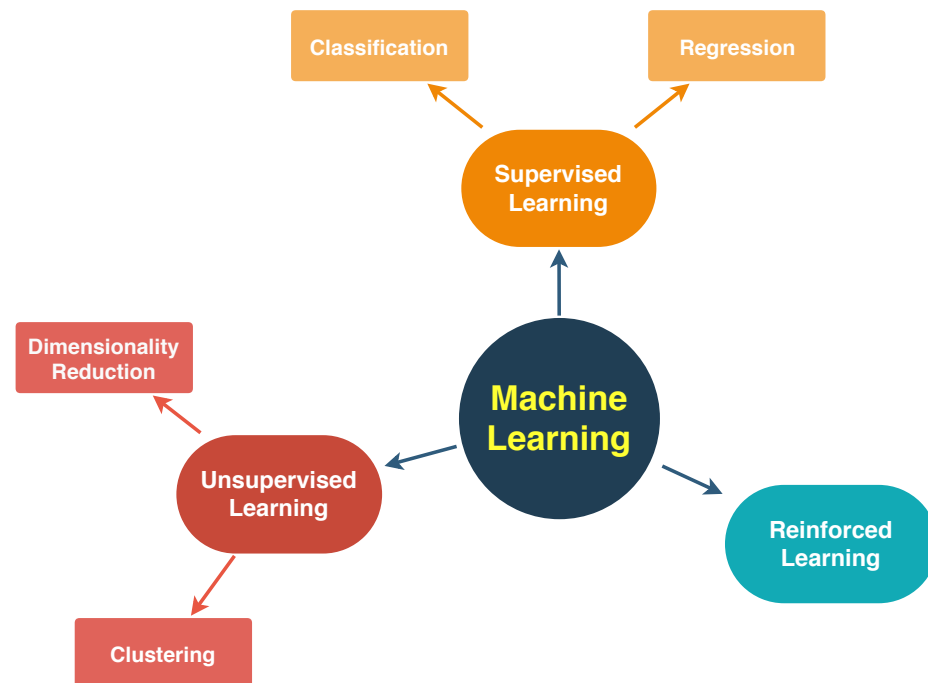
1. Autoencoders
2. Labs, hand on practice

Machine Learning subcategories:

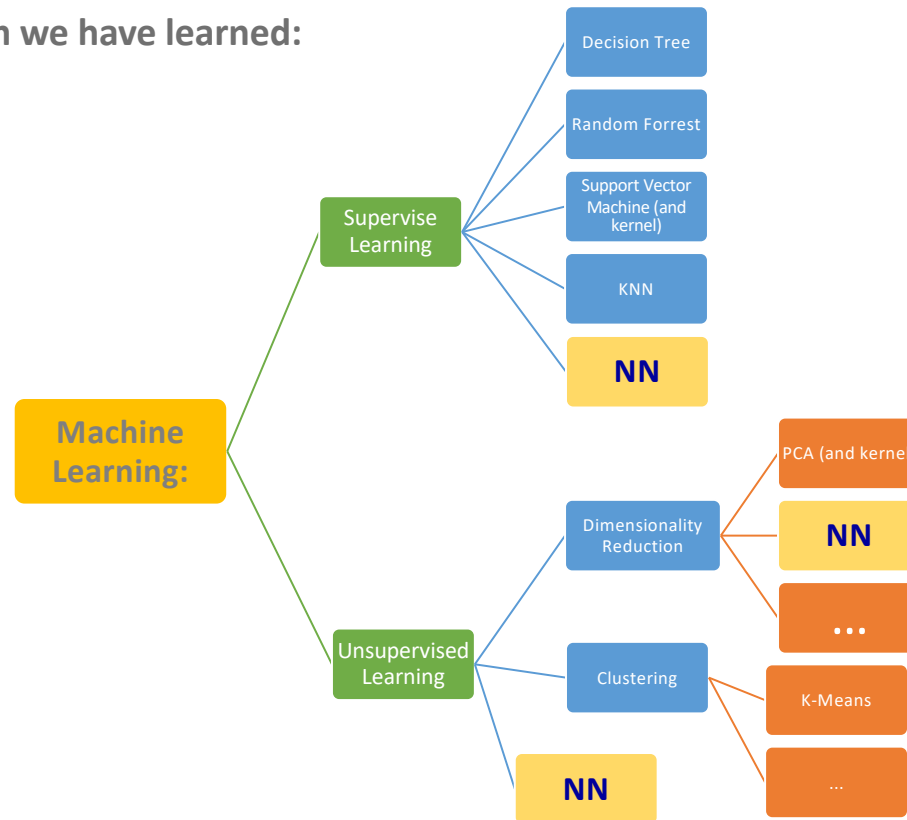
- I. Supervised Learning
- II. Un-supervised Learning
- III. Reinforced learning
- IV. Others...

Where is NN:

Almost all of them



What algorithm we have learned:

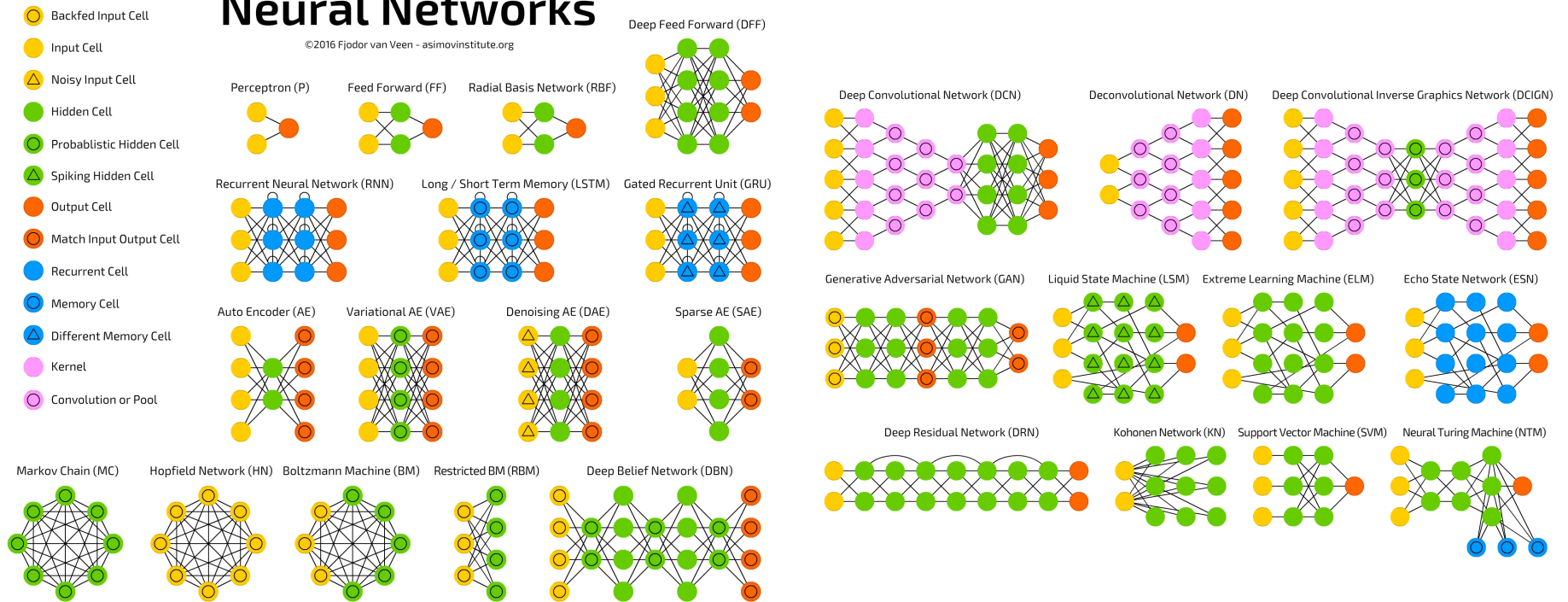


Different structures of Neural Networks:

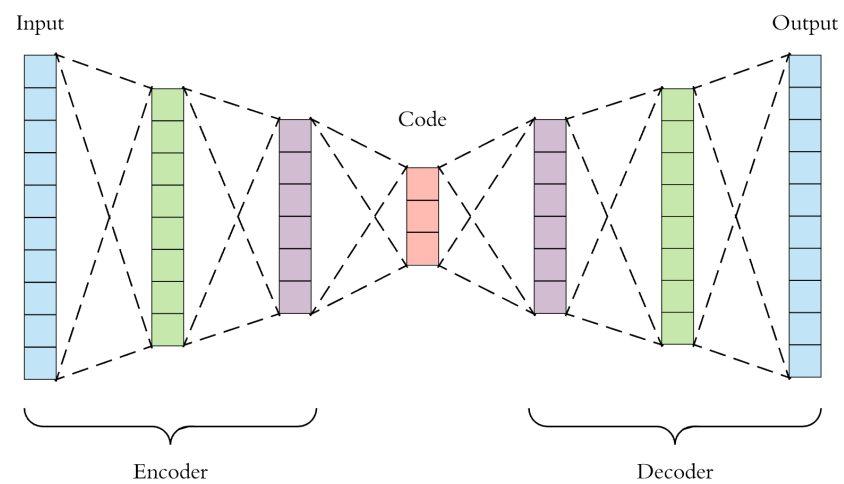
A mostly complete chart of

Neural Networks

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Autoencoders

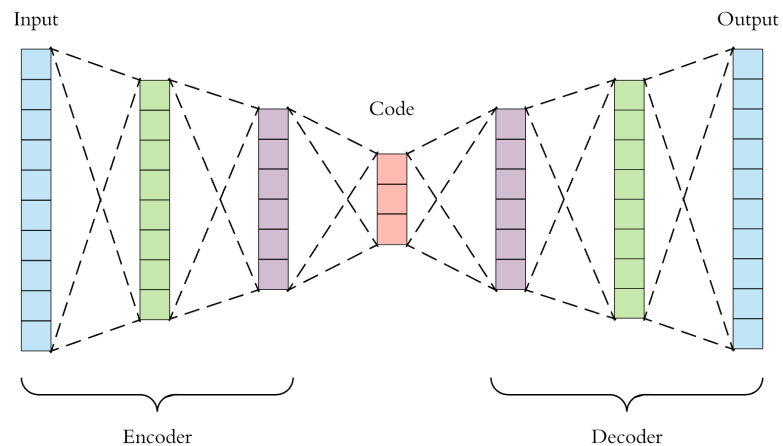


What are Autoencoders:

As the name suggest it is a type of NN that encode data, by provide a representation of data that are more important (i.e remove what is not important).

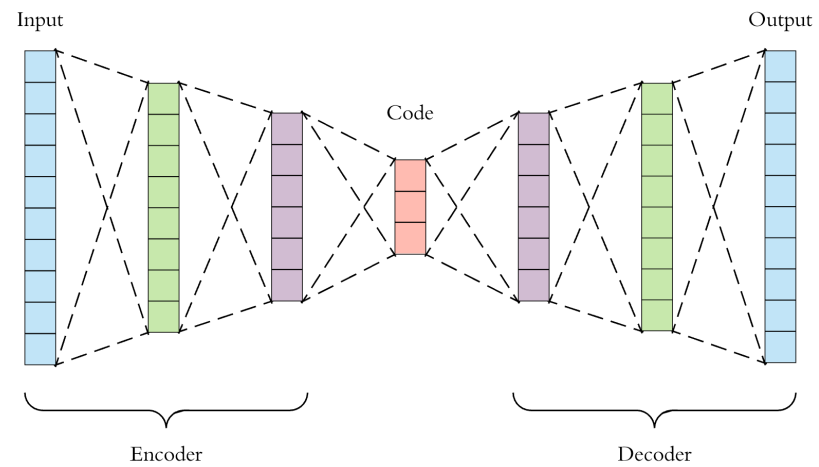
It can be also think as a dimensionality reduction algorithm. At the network bottleneck data is represented in a compressed form.

Autoencoders are often classified as unsupervised learning algorithm but can also be used in a supervised format.



Autoencoders

The Autoencoder will encode data (encoder) and then decode it back (decoder) to create the input.



Applications of Autoencoders:

Dimensionality Reduction

recommenders

Image Compression

Image Denoising

Feature Extraction

Image generation

Note that different type of NN like RNN or CNN can be build in form of an Autoencoders.



([Wojciech Mormul on Github](#))

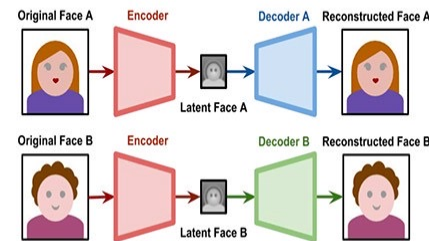


Image Source: <https://www.alanzucconi.com/2018/03/14/understanding-the-technology-behind-deepfakes>

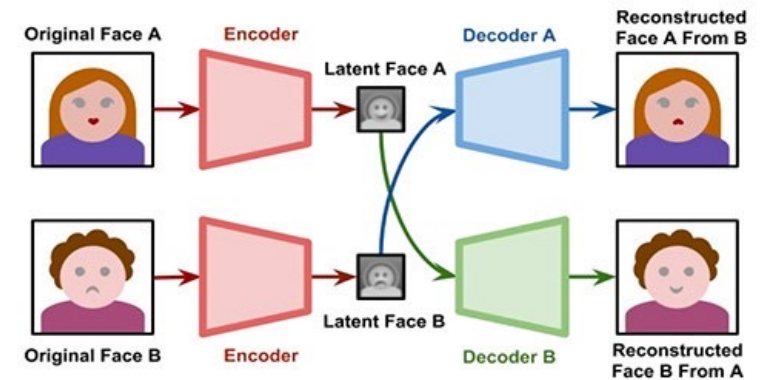
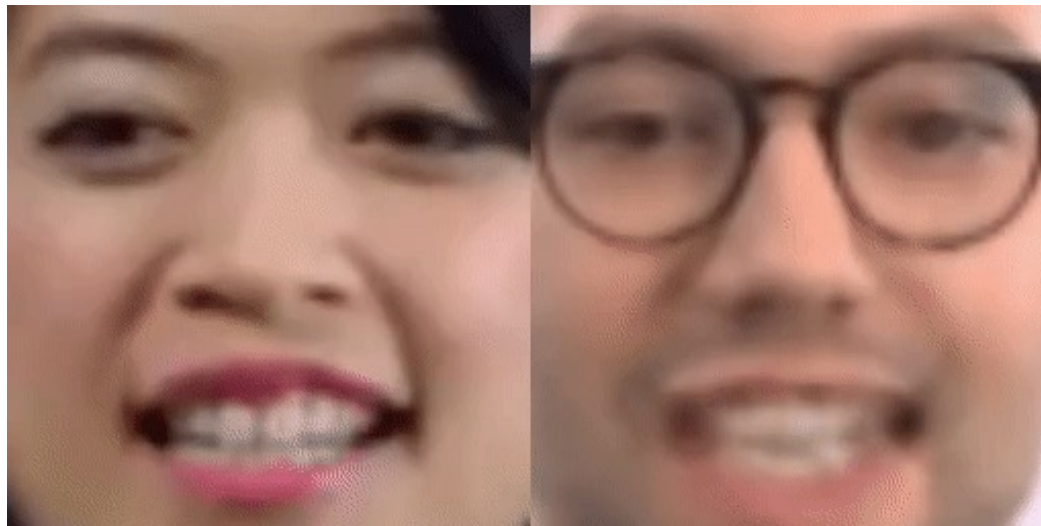


Image Source: <https://www.alanzucconi.com/2018/03/14/understanding-the-technology-behind-deepfakes>

Applications of Autoencoders:

Fun applications:

<https://www.youtube.com/watch?v=LeeBnlZLTBs>



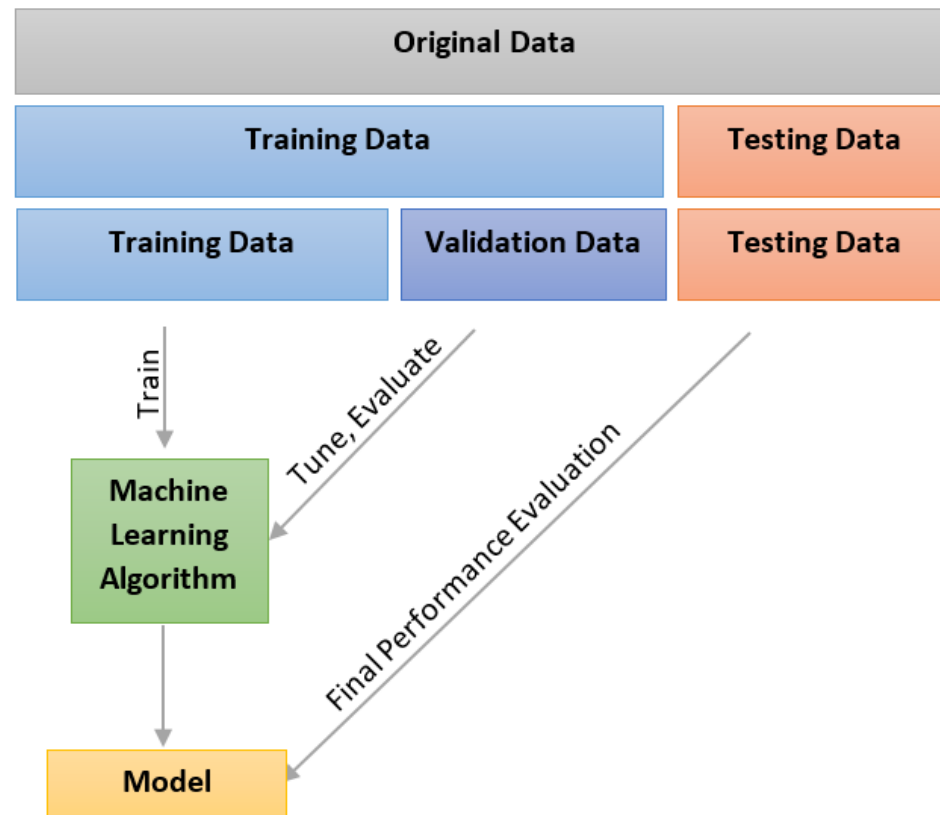
Python:



Denoising images using Autoencoders and CNN Autoencoders

Training vs Test sets:

- 1) Split dataset into Train and test sets
- 2) Train model on the training set
- 3) Tune model parameters on Validation set
- 4) Perform final evaluation on test set





Python:

Scikit learn: Linear Regression

For almost every model we do, we follow these:



Sklearn docs:

https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LinearRegression.html

Python:



Scikit learn:

A free library in Python that has great features for clustering, classification, regression and dimensionality reduction.

scikit-learn
algorithm cheat-sheet

